UNIVERSITÄT BAYREUTH

Modelling Seminar Topic Presentation 2026

T. Rosnitschek

Lehrstuhl für
Konstruktionslehre und CAD
Prof. Dr.-Ing. Stephan Tremmel

STL – Part conversion using LLM-based Reverse Look Up

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Issue:

- Topology-Optimization generates meshed, respectively STL parts
- Those mesh-based Design Proposals cannot be used in CAD-environment for fine-tuning and manufacturing preparation
- Existing Geometry Converter result in highly fragmented or randomly segmented facets, which are also hardly usable for CAD-Designer





Idea:

- Analyse the CAD-Sequence of parts
- Use LLM to generate CAD-Sequences for given STL parts



MISTRAL AI_

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What is to do?

- Design 20 heterogeneous parts in FreeCAD (between 5 and 20 CAD-Operations)
- Use the Python-Interface to generate ca. 100 Variants of each part, Save CAD-file, FeeCAD-Python Script and STL-Mesh of each part
- Those are your data, while STL is the input, the FreeCAD-Script Output and the CAD-File the ground-truth
- Fine-tune LLM to generate CAD-files via FreeCAD-Scripts based on the STL-file
- Use Shape-Optimization for improving the AI-result





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What is to investigate?

- How does the number of CAD-operations influence the quality of the result?
- How can system prompts improve the performance of the workflow and how robust is it for prompting variations?
- Vary your Fine-Tune parameter (LoRa etc.) systematically and investigate their influence
- How to automatically extract the parameter from the generated CAD-part for the shape Optimization?
- Use Shape-Optimization for improving the AI-result





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Starting point is an existing GitLab project with:

- Previous Work on LLM-based generation of bolt connections in CAD
- Last Years report and Code



References:

- Journal Paper: https://doi.org/10.1016/j.cad.2026.104065
- Special Issue: https://www.sciencedirect.com/special-issue/10MLBP6Q2LP
- Zenodo Repository with the code: https://doi.org/10.5281/zenodo.18269442